

21	B	increase in K.E. = decrease in E.P.E. $\frac{1}{2}mv^2 - 0 = q \Delta V $ $v = \sqrt{\frac{2q \Delta V }{m}} = \sqrt{\frac{2qEd}{m}} \Rightarrow v \propto \sqrt{Ed} \text{ for the same charge to mass ratio}$ $\frac{v_2}{v_1} = \sqrt{\frac{E_2 d_2}{E_1 d_1}}$ $v_2 = \left(\sqrt{\frac{2E(2d)}{Ed}} \right) v$ $= 2v$

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22	C	$I_1 = I_2 = I_3$
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